

Statistical Analysis of the Evolution of Professional Tennis

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Introduction

Problem Description

The purpose of this study is to investigate how professional tennis has evolved over time by analyzing historical match data from ATP and WTA tournaments. The sport has experienced substantial changes driven by technological developments, advances in training methodologies, shifts in tactical approaches, and changes in player physiology. Using a comprehensive dataset that includes match statistics such as aces, serve efficiency, match duration, as well as player characteristics including height, age, and ranking, this study seeks to uncover significant temporal trends and structural transformations within professional tennis.

The central research question of interest is:

How has tennis changed over time in terms of:

- player characteristics [Reguş Albert],
- match performance metrics (ATP) [Raicu Maria], and
- gender-based differences between ATP and WTA tournaments [Roşca Mitruţ]

Objectives

What is to Be Studied

The analysis focuses on identifying:

- **Evolution of Serving Dominance:** Investigating whether the average number of aces per match has changed significantly across decades (1990s–2020s) and confirming if these shifts are statistically significant.
- **Variability and Predictability:** Assessing whether the variance in match duration and serving performance has increased over time, indicating a widening gap between playing styles and reduced match predictability.
- **Global Shifts in Talent:** Analyzing the geographic distribution of Top 100 players to identify significant shifts in regional dominance, specifically comparing the USA, Europe, and the Rest of the World between the 1990s and 2010s.
- **Physical Pace of Play:** Evaluating the "density" of the game by measuring the trend in points played per minute, to determine if the sport has physically slowed down or accelerated over the last 30 years.
- **Comparison of growth trends:** Investigate the evolution of player physiology to see growth patterns and differences over decades.
- **Evolution of handedness:** Investigating if the handedness changed over time and whether it's significant
- **Handedness on Aces::** Analyzing the relation between the hand of the player and ace counts to observe if handedness affects serving performance.
- **Differences in Serve Effectiveness:** Comparison between the number of aces served in the ATP (men's) versus the WTA (women's) to observe how they differ structurally.
- **Impact of Height on Serving:** Analyzing the relationship between height and ace counts to see if being tall provides a different level of advantage in the ATP compared to the WTA.
- **Competitiveness and Match Length:** Testing the theory that because serve dominance is lower in the WTA (leading to more break points), women's matches are more likely to go to a deciding set.

Expected Outcomes

We aim to determine whether modern tennis is more serve-dominant, physically demanding, or competitive than in previous decades. Through the identification of long-term trends and statistically significant relationships, this analysis is expected to highlight structural changes in match dynamics, player physiology, and performance indicators across eras.

Data Collection

The dataset used in this project originates from the repository titled *“Tennis Rankings, Results, and Stats”*¹, compiled from historical match records of professional tennis. It includes detailed match-level statistics such as serve performance, break points, match duration, player demographics, rankings, and tournament metadata. The dataset contains hundreds of thousands of records spanning multiple decades and both the men’s (ATP) and women’s (WTA) tours.

Descriptive Statistics

Descriptive statistics were used to obtain an overview of the main variables. Various graphical tools were used for visualization:

- **Histograms** were used to examine the distribution of winner aces (Figure 1a).
- **Boxplots** were used to compare Match Duration across eras (Figure 1b).
- **Trend Lines** were used to display the percentage of matches going to a deciding set over time (Figure 1c).
- **Scatter Plots** were used to visualize aces versus height (with linear regression line) (Figure 1d), height over time (with linear regression line) (Figure 1e; and the distribution of handedness over time (Figure 1f).

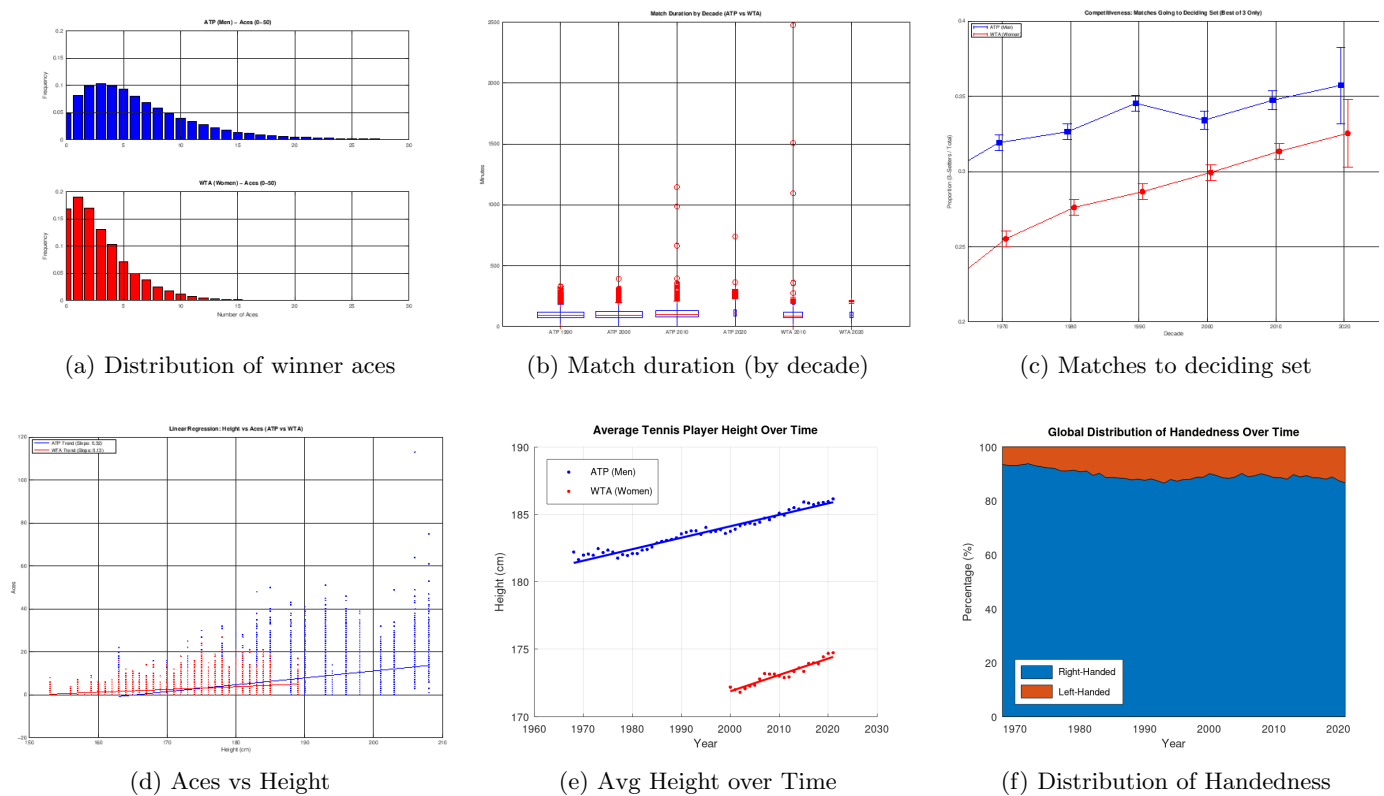


Figure 1: Comprehensive analysis of tennis metrics. The top row displays distribution and competitiveness, while the bottom row focuses on player physical attributes.

Inferences

Statistical Methods

A variety of inferential techniques were applied:

¹<https://github.com/JeffSackmann>

- **Confidence intervals** for mean aces per match for each decade.
- **T-tests** to compare performance indicators between time periods or surfaces and between handedness and aces.
- **F-tests** to compare variances in match length between time periods.
- **Chi-square tests** to see if there are regions that produces more professionals players now compared to the past and if the handedness of the players changed over time.
- **Correlation analysis** to measure relationships between height and aces, minutes and total points.
- **Linear regression** to quantify temporal trends, such as the increase in aces, changes in match duration over time and to analyse growth trends in tennis players.

Results and Interpretation

Variables	Method		Result	Interpretation
Aces per Match (Decadal)	95% CI	for	Mean increase from 5.68 (90s) to 7.40 (20s) with non-overlapping intervals.	Statistically significant increase in serving dominance over time.
Aces (1990s vs 2010s)	Welch's T-Test		$p < 0.001$, $t = -34.63$. Mean increase by ≈ 1.5 aces.	Difference is statistically significant; confirms serving evolution hypothesis.
Variance (Duration & Aces)	F-Test for Variances		Variances increased by $\approx 47\%$ ($p < 0.001$) for both metrics.	Predictability has dropped; the gap between distinct playing styles has widened.
Regional Dominance (Top 100)	Chi-Square Test		$p < 0.001$, $\chi^2 \approx 1456$. USA share fell from 18.5% to 7.1%.	Significant geographic shift: USA dominance collapsed while Europe's share grew to $\approx 68\%$.
Pace of Play (Pts vs Min)	Linear	Regression	Slope dropped from 1.42 (90s) to 1.17 (20s) pts/min.	The game has slowed down physically; fewer points are played per hour of broadcast.
Height over time (Height vs Year)	Linear	Regression	ATP slope: +0.085; WTA slope: +0.1214.	Men are growing taller by 0.85 cm/decade, while women are growing at a faster rate of 1.21 cm/decade.
Handedness over time (Handedness vs Decade)	Chi-Square Test		$\chi^2 = 73.0$, $p < 0.001$; Left-handers increased from 7.7% (1970s) to 12.8% (2020s).	The distribution of handedness has changed significantly; left-handed players have become increasingly prevalent in the modern game compared to previous decades.
Handedness on Aces (ATP) (Ace Rate: L vs R)	Welch's T-test		$t = -2.92$, $p = 0.0035$; Righties (7.21%) ; Lefties (7.09%).	Right-handers serve aces at a higher rate.
Handedness on Aces (WTA) (Ace Rate: L vs R)	Welch's T-test		$t = 2.34$, $p = 0.019$; Lefties (3.85%) ; Righties (3.72%).	Left-handers serve aces at a higher rate.
Aces (ATP vs WTA)	Welch's T-test		$t = 115.25$, $p < 0.0001$	Men hit $2.14\times$ more aces per match
Height on Aces (ATP vs WTA)	Linear	Regression	$ATP_{slope} : +0.32$; $WTA_{slope} : +0.13$	ATP: 10cm would add 3.2 aces per match ; WTA: 10cm would add 1.3 aces per match
Match Competitiveness (ATP vs WTA)	95% CI for proportion		ATP: 34.2% ; WTA: 30%	Men play deciding sets more often (around 4.2%)

Table 1: Key findings on the evolution of tennis match performance over time

Conclusions

This study confirms that professional tennis has undergone significant structural and physiological changes over the past decades. We observed a statistically significant rise in serve dominance, strongly correlated with increased player height in both tours. Paradoxically, while serving power has peaked, the physical pace of play (points per minute) has slowed. Demographically, the sport has shifted from American to European hegemony, and the prevalence of left-handed players has nearly doubled. Finally, comparisons reveal that while the ATP is more serve-oriented, it also features a higher frequency of deciding sets than the WTA, indicating distinct competitive dynamics between the tours.